A

Seminar Report

On

"CRISPR Gene Editing Technology"

Submitted for partial fulfilment of the requirement for the degree of

BACHELOR OF ENGINEERING

(Computer Science & Engineering)

Submitted By

Mr. Akash Wagh

Under the Guidance of

Prof. Akhil. M. Jaiswal

Department of Computer Science & Engineering

Hanuman Vyayam Prasarak Mandal's College of Engineering & Technology, Amravati.

Sant Gadge Baba Amravati University, Amravati.

Year 2025-2026

Certificate

This is to certify that the Seminar entitled

"CRISPR Gene Editing Technology"

Is a bonafide work and it is submitted to the Sant Gadge Baba Amravati University, Amravati.

By

Mr. Akash Wagh

For the partial fulfilment of the requirement for the degree of Bachelor of Engineering in Computer Science & Engineering, during the academic year 2025-2026

Prof. Akhil. M. Jaiswal
Guide
Dept. of Computer Science & Engineering

Dr. A. B. Raut
Head
Dept. of Computer Science & Engineering

Department of Computer Science & Engineering

Hanuman Vyayam Prasarak Mandal's
College of Engineering & Technology, Amravati.
Sant Gadge Baba Amravati University, Amravati.
Year 2025-2026

INDEX

Sr.no	ABSTRACT	Pg No
1	INTRODUCTION	1
2	LITERATURE REVIEW	3
3	TRADITIONAL VS CRISPR	4
4	COMPONENTS AND MECHANISM	5
5	WORKING	7
6	APPLICATION	12
7	ADVANTAGE AND DISADVANTAGE	13
8	FUTURE SCOPE	14
9	CONCLUSION	15
	REFRENCES	16

ACKNOWLEDGEMENT

It is a matter of great pleasure to get the opportunity to highlight afraction of the knowledge I acquired during my technical education through this Seminar.
This would not have been possible without the guidance and help of many people. This is the only page where I have the opportunity to express my emotions and gratitude from the core of my heart to them.
This Seminar would not have been successful without enlightened ideas timely suggestions and the keen interest of my respected Guide **Prof. Akhil.M.Jaiswal** the best guidance this would have been an impossible task to complete.
Being on the same line I express my deep sense of gratitude to our Head of Department **Dr. A. B. Raut** for the most valuable guidance provided by her. Last but not least; I would like to express my thankfulness to the teaching and non-teaching staff, my friends and all my well-wishers.

Mr. Akash Wagh

Final Year Computer Science & Engineering,
HVPM's College of Engineering and Technology, Amravati

ABSTRACT

CRISPR-Cas9 is a revolutionary gene-editing technology that enables precise targeting and modification of DNA sequences. Originally discovered as a bacterial immune defense mechanism, it has been adapted into a powerful tool for biotechnology. The system uses a guide RNA and the Cas9 enzyme to cut DNA at specific sites, allowing insertion, deletion, or correction of genes with remarkable accuracy. This technology has shown great promise in medicine for curing genetic disorders, treating cancer, and combating infectious diseases. In agriculture, it is applied to enhance crop yield, resistance to pests, and nutritional value, contributing to food security. Beyond health and farming, CRISPR is also explored for conservation, de-extinction projects, and biodiversity management. Its industrial applications include biofuel production, enzyme synthesis, and environmental bioremediation. Compared to traditional methods like ZFNs and TALENs, CRISPR is simpler, faster, cost-effective, and more efficient. Despite its vast potential, challenges such as off-target effects, delivery methods, ethical debates, and regulatory restrictions remain. Overall, CRISPR holds immense potential to transform science, medicine, agriculture, and sustainability in the coming future.

1. INTRODUCTION

CRISPR, which stands for Clustered Regularly Interspaced Short Palindromic Repeats, is an advanced gene-editing technology that allows scientists to precisely modify DNA sequences. It was adapted from a natural bacterial immune system where CRISPR and Cas proteins defend against viruses. The CRISPR-Cas9 system uses a guide RNA to identify a target DNA region and the Cas9 enzyme to cut it, enabling the addition, removal, or correction of genes [1][2].

In medicine, CRISPR has shown great promise in treating genetic diseases such as sickle cell anemia and β-thalassemia, and in developing new therapies for cancer and viral infections. Researchers are also exploring its potential for personalized medicine, where treatments can be tailored to an individual's genetic makeup. Clinical trials have demonstrated encouraging results, making CRISPR a powerful tool for next-generation healthcare [6][7].

In agriculture, CRISPR is helping to develop crops with improved yield, pest resistance, and environmental tolerance. Examples include non-browning mushrooms, drought-resistant maize, and disease-resistant rice varieties. These innovations contribute to food security and sustainable farming by reducing the need for pesticides and improving crop nutrition [3][4][5].

CRISPR also supports **de-extinction and bioinformatics research**. Scientists are using it to bring back traits of extinct species such as the woolly mammoth by editing the DNA of closely related animals like Asian elephants [8]. Meanwhile, bioinformatics provides essential c**omputational** tools to design accurate guide RNAs, identify gene targets, and minimize off-target effects, ensuring safer and more effective genome editing. Together, these technologies are transforming modern biotechnology across medicine, agriculture, and conservation.

In industrial and **environmental biotechnology**, CRISPR is being used to engineer microorganisms that produce biofuels, useful enzymes, and biodegradable materials. It helps create eco-friendly solutions for reducing industrial waste, improving biofuel efficiency, and supporting a cleaner environment. By integrating CRISPR with green technology, scientists aim to achieve sustainable industrial development while reducing the effects of climate change.

CRISPR Technology

CRISPR is a powerful tool for genome editing that enables researchers to quickly alter DNA sequences and alter how genes function. It offers a wide range of possible uses, including the correction of inheritable diseases, the treatment and prevention of illness, and the enhancement of crop growth and adaptability.

Cas9, also known as "CRISPR-associated protein 9," is an enzyme that recognizes and cleaves particular DNA strands that are complementary to CRISPR sequences by using the CRISPR sequences as a guide. The CRISPR-Cas9 technique can be used to modify the genes of organisms by combining Cas9 enzymes with CRISPR sequences. The production of biotechnological products, the treatment of diseases, and basic biological research are just a few of the many uses for this editing.

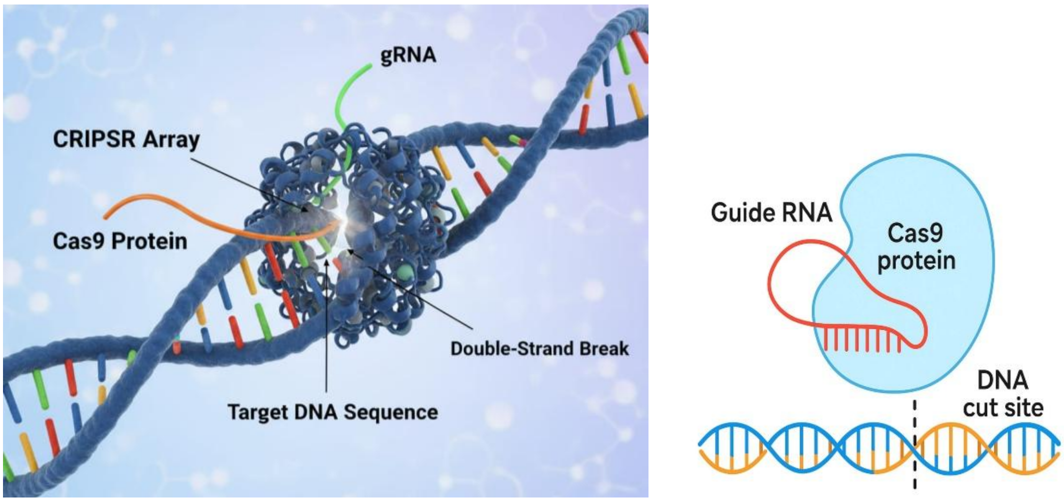

Fig. 1.1 CRISPR Technology

Key Features of CRISPR-Cas9

- **High Precision** – Targets specific DNA sequences with the help of guide RNA.
- **Cost-Effective** – Much cheaper compared to older gene-editing tools like ZFNs and TALENs.
- **Speed** – Allows gene editing in days or weeks instead of years.
- **Versatility** – Can be used in plants, animals, and humans for different applications.
- **Programmable** – Guide RNA can be easily designed for almost any DNA sequence.

2.LITERATURE REVIEW

Gene editing has developed significantly over the years. Early methods like selective breeding, cloning, and mutagenesis were used to improve plant and animal traits. These methods were slow, labor-intensive, and often inaccurate, which made it difficult to achieve precise results quickly.
Later, ZFNs (Zinc Finger Nucleases) allowed scientists to target specific genes more precisely. Although more accurate than traditional methods, ZFNs were expensive and technically complex, limiting their use to specialized labs and research projects.

The discovery of CRISPR-Cas9 revolutionized gene editing. It is fast, accurate, and easy to use, making it a powerful tool in agriculture, medicine, and conservation. In agriculture, CRISPR has been used to develop disease-resistant and high-yield crops. In medicine, it helps treat genetic disorders and study gene functions. It is also applied in conservation efforts, such as protecting endangered species or exploring de-extinction.
Recent studies show CRISPR's transformative impact worldwide. Experiments in crops like rice and wheat have demonstrated improved growth, yield, and disease resistance, while medical research focuses on correcting faulty human genes. Scientists continue to study off-target effects and ethical concerns to ensure safe and responsible use.

3.TRADITIONAL VS CRISPR

Traditional breeding has been used for centuries to improve plants and animals. It involves crossbreeding individuals with desirable traits over multiple generations or using older genetic modification techniques. While effective, this method is slow and often imprecise, as it mixes thousands of genes, which can sometimes introduce unwanted traits. Traditional methods are also limited to species that can naturally interbreed, and achieving significant improvements can take years or even decades.

CRISPR-Cas9 technology, on the other hand, is a modern gene editing tool that allows scientists to directly edit DNA at precise locations. Using the Cas9 enzyme, guided by RNA, specific genes can be added, removed, or corrected. This makes CRISPR much more precise than traditional methods, reducing the risk of unintended changes. It is also faster, often producing desired results in weeks or months, rather than generations.

The applications of these two approaches also differ. Traditional methods are mainly used for crop improvement, livestock breeding, and older formsof genetic research, while CRISPR has broader uses, including medicine (gene therapy), disease-resistant crops, and even de-extinction projects.While CRISPR is cost-effective and efficient, it also raises ethical concerns, such as ecological impact or the possibility of designer genes. In summary, traditional methods are slower and less precise, whereas CRISPR is revolutionary, fast, and highly targeted, offering new possibilities in genetics and biotechnology

Crossbreeding – “Quagga Project”

Cloning – “Dolly the Sheep”

ZFN – Zinc Finger Nucleases

Fig. 1.2 Traditional method

4.COMPONENTS AND MECHANISM

The CRISPR-Cas9 system is a powerful and precise gene-editing tool derived fro natural bacterial immune system. It enables targeted changes in DNA and has two main components**:**

- **Cas9 Enzyme:** Acts as molecular "scissors" that cut DNA. Cas9 is a nuclease enzyme, meaning it cleaves the phosphodiester bonds of the DNA backbone**.**
- **Guide RNA (gRNA**): The guide RNA (gRNA) is a critical component of the CRISPR-Cas9 system that directs the Cas9 enzyme to the exact location in the DNA that needs to be edited.

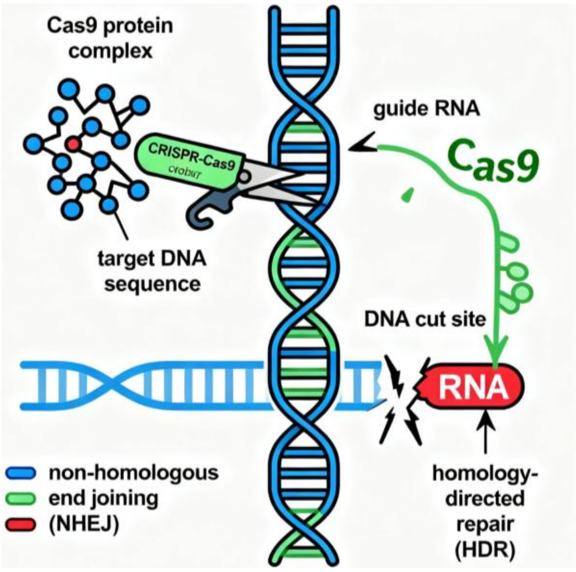

Fig 1.3 COMPONENTS AND MECHANISM

Mechanism of Action of CRISPR-Cas9

The CRISPR-Cas9 system works through a precise and targeted process to edit genes. Its mechanism of action involves the following steps:

- **Formation of Cas9-gRNA Complex:** Scientists design a **guide RNA (gRNA)** that matches the target DNA sequence. The gRNA binds to the Cas9 enzyme, forming a stable Cas9-gRNA complex.
- **Target Recognition:** The complex scans the genome, and the gRNA recognizes the target DNA sequence through complementary base pairing.
- **PAM Sequence Check:** For Cas9 to cut the DNA, the target site must be next to a **Protospacer Adjacent Motif (PAM)**, usually "NGG." Cas9 only becomes active when it detects this PAM sequence adjacent to the target.

- **DNA Cleavage:** Once positioned correctly, Cas9 acts as molecular scissors to make a **double-stranded break (DSB)** in the DNA, usually three base pairs upstream of the PAM site.
- **DNA Repair and Editing:** The cell's natural repair systems fix the break,and scientists exploit these pathways to achieve gene editing:
    - **Non-Homologous End Joining (NHEJ):** A quick but error-prone repair that may introduce small insertions or deletions, often inactivating the gene.
    - **Homology-Directed Repair (HDR):** A precise repair method using a **donor DNA template** to insert a new sequence or correct a mutation exactly at the break site.
- **Result:** The targeted gene is successfully **added, removed, or modified** with high precision, enabling applications in medicine, agriculture, and research.

5.WORKING

i. Working in Agriculture

CRISPR helps farmers grow better, stronger, and more nutritious crops by precisely editing plant genes. This process is faster and more accurate than traditional cross-breeding.

Step 1: Identify the Problem in the Crop.

Scientists first **observe and analyze the crop** to identify issues affecting affecting its growth or quality.

Examples of problems:

- Susceptibility to diseases or pests (e.g., wheat fungus, rice blast).
- Low yield or slow growth.
- Short shelf life of fruits and vegetables.
- Lack of essential nutrients, such as vitamins or minerals.

Goal: Define a clear trait that needs improvement.

Step 2: Select the Target Gene.

- Every trait in a plant is controlled by specific genes in its DNA.
- Scientists use genetic databases and bioinformatics tools to:
- Locate the exact gene responsible for the problem.
- Study its sequence and role in the plant**.**

Example:

- A gene that causes tomato softening quickly after harve.
- A gene that makes rice plants sensitive to salty soil.

Step 3: Design the Guide RNA (gRNA).

- Guide RNA (gRNA) acts like a GPS to direct the CRISPR system to the correct spot in the DNA.
- It is carefully designed to:
- Match the target DNA sequence perfectly.
- Avoid cutting unintended parts of the plant genom(off-target effects).

Step 4: Cutting the DNA.

- Once inside the plant cell:
- The **gRNA finds the target gene** using its matching sequence.
- Cas9 then **cuts the DNA precisely at that location**, similar to scissors cutting a string.
- This cut creates a **double-stranded break** in the DNA.

Step 5: Plant DNA Repair Mechanisms.

a) NHEJ (Non-Homologous End Joining)

- The plant reconnects the DNA ends directly.
- This often introduces **small errors**, which can:
- Turn the gene **off** or **disable it**.
- Useful for removing unwanted traits like disease sensitivity or fast

b) HDR (Homology Directed Repair)

Scientists provide a **template DNA strand** during the repair process.

- The plant uses this template to **insert or replace the gene** accurately.
- Useful for adding **beneficial traits**, such as:
  - Drought tolerance.
  - Increased nutritional value.
  - Pest resistance

Step 6: Regenerating the Edited Plant

- These plants now **carry the improved trait** permanently in their DNA
- The edited cells are **grown in controlled lab conditions** to form a complete plant.

Step 7: Field Testing and Validation

- The new plants are tested in greenhouses and fields to ensure:
  - Desired traits are stable.
  - No harmful effects on the environment or human health.
  - Improved performance under real agricultural conditions.

Step 8: Improved Crop is Ready

- The final crop now has **enhanced characteristics**, such as:
  - **Higher yield** for increased food production.
  - **Natural resistance** to diseases and pests.
  - **Improved climate resilience**, surviving drought or extreme heat.
  - **Better nutritional value** with added vitamins or minerals.
  - **Extended shelf life**, reducing food spoilage and waste.

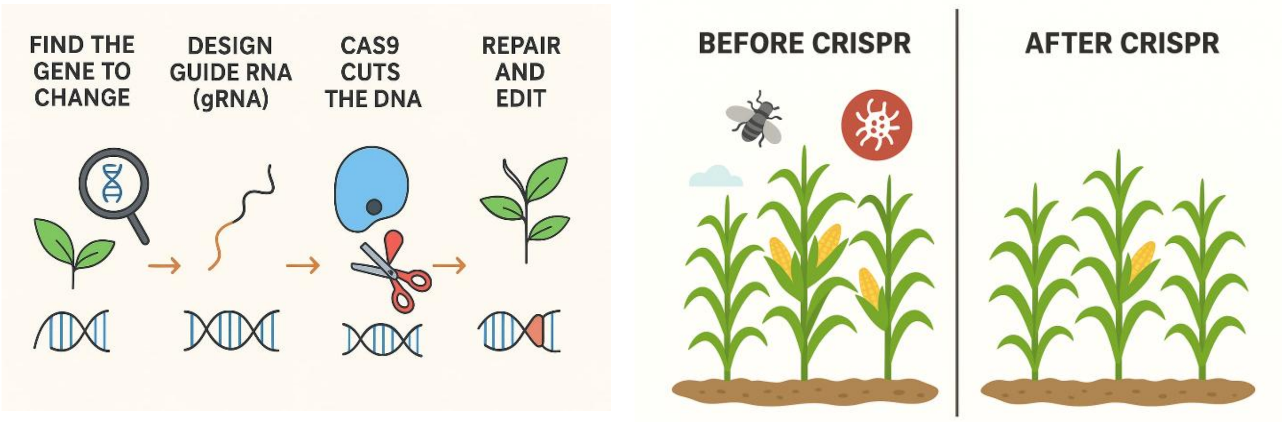

Fig. 1.4 CRISPR Technology in agriculture

+

ii. De-Extinction

Step 1:Identify the Extinct Species

- Scientists choose a species to revive, such as the **woolly mammoth**, **passenger pigeon**, or **dire wolf**.

Step 2: Collect DNA Samples

- DNA is extracted from fossils or preserved remains like bones, teeth, or frozen tissue.

Step 3: Collect DNA Samples

- Find a **closely related living species** (e.g., elephants for woolly mammoths, modern wolves for dire wolves).
- Their DNA is compared to identify missing or damaged genes.

Step 4: Compare with a Close Living Relative

- CRISPR is used to **modify the living relative's DNA** to match the extinct species' genes.
- Example: Adding mammoth genes for thick fur and fat layers into elephant DNA.

Step 5: Insert Edited DNA into an Egg Cell

- The modified DNA is placed into an egg cell and then implanted into a **surrogate mother** of the related species.

Step 6: Birth of the New Animal

- The surrogate gives birth to an animal that **resembles the extinct species** genetically and physically.

Step 7: Reintroduce to the Environment

- After survival and behavior testing, these animals are released into protected habitats to restore ecosystems.

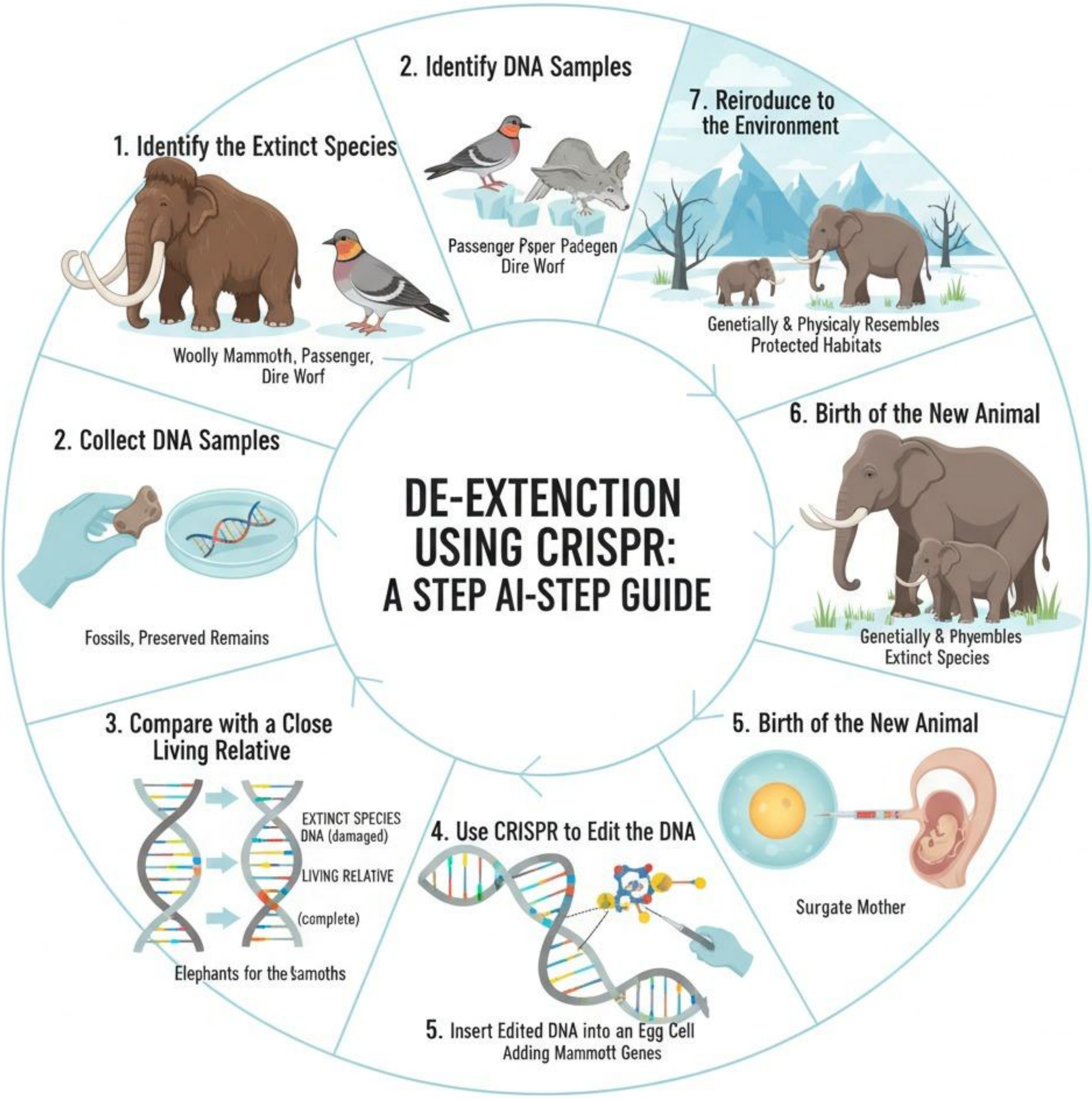

Fig. 1.5 CRISPR Technology in De-Extinction

BIOFORMATICS

Bioinformatics is the application of computational tools, algorithms, and databases to analyze and interpret biological data such as DNA, RNA, and protein sequences. It integrates biology, computer science, and mathematics to solve complex biological problems.

Bioinformatics plays a critical role in CRISPR technology by helping scientists:

1. **Analyze DNA sequences** – Identify the best target genes or regions for editing.
2. **Design guide RNAs (gRNAs)** – Predict which gRNAs will efficiently and accurately target the desired DNA sequence.
3. **Predict off-target effects** – Scan the genome for similar sequences to prevent unintended edits.
4. **Plan experiments** – Provide protocols and guidance for synthesizing gRNAs and performing CRISPR edits.
5. **Interpret results** – Analyze sequencing data to confirm successful gene modifications

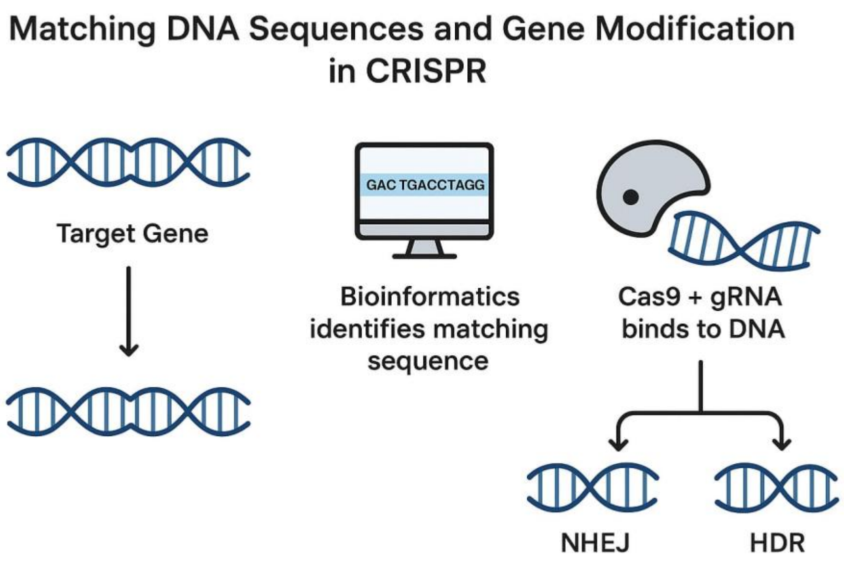

Fig. 1.6 Bioinformatics CRISPR Technology

6.APPLICATION

1.Medicine & Healthcare

- Gene Therapy: Corrects genetic disorders by editing faulty genes (e.g., sickle cell anemia, cystic fibrosis).
- Cancer Treatment: Targets and modifies immune cells to fight cancer more effectively (CAR-T therapy enhancements).
- Infectious Disease Resistance: Engineering resistance to viruses like HIV or hepatitis.
- Drug Development: Helps create cell or animal models with specific mutations to test drugs.

2. Agriculture

- Crop Improvement: Enhances yield, nutritional content, and drought or pest resistance.
- Livestock Enhancement: Improves disease resistance, growth rate, or meat quality in animals.
- Reduced Pesticide Use: Plants can be edited to naturally resist pests, reducing chemical usage.

3. Environmental Applications

- De-extinction: Potentially revive extinct species or protect endangered ones by correcting genetic issues.
- Conservation: Enhance genetic diversity in endangered populations.
- Bioremediation: Engineer microbes to clean up pollutants or toxic waste.

4. Industrial Biotechnology

- Biofuel Production: Engineer microorganisms for more efficient biofuel synthesis.
- Enzyme Production: Create microbes that produce industrially important enzymes more efficiently.
- Synthetic Biology: Design custom organisms for specific industrial purposes.

7.ADVANTAGE AND DISADVANTAGE

Advantages

One of the major advantages of CRISPR-Cas9 is its high precision, where the guide RNA directs Cas9 to cut DNA at a specific site, allowing accurate changes in the genome . Compared to earlier methods such as Zinc Finger Nucleases (ZFNs) and TALENs, CRISPR is faster, cheaper, and easier to design, making it more accessible for scientists worldwide. It is also highly versatile, with applications in agriculture, medicine, and conservation. For example, it has been used to produce disease-resistant crops improve staple foods like rice and maize and even develop non-browning mushrooms. In medicine, CRISPR has shown success in treating genetic diseases like sickle cell anemia and β-thalassemia while cancer therapies using CRISPR-modified immune cells are also under clinical testing . In addition, its ability to perform multiplexing, or editing multiple genes at once, gives it a significant advantage over traditional techniques.

Disadvantages

The most common concern is off-target effects, where the tool may cut DNA at unintended locations, leading to harmful mutations. Another limitation is the difficulty of delivering CRISPR safely into cells and tissues, especially in humans. Beyond technical issues, the technology raises serious ethical concerns, particularly in human embryo editing, germline modifications, and controversial projects like de-extinction. Scientists also caution that the long-term impacts of CRISPR edits are not fully known, which makes its widespread use in medicine risky. Moreover, governments in many countries enforce strict regulations on CRISPR applications in food, healthcare, and biodiversity, slowing down its adoption.

8. FUTURE SCOPE

1. Precision Medicine

- Personalized treatments designed for each patient's unique genetic makeup.
- Permanent cures for genetic disorders like sickle cell anemia, hemophilia, and muscular dystrophy.
- Early detection and prevention of genetic diseases before birth.

2. Advanced Agriculture

- Climate-smart crops capable of thriving under extreme weather conditions like drought or floods.
- Completely pesticide-free farming through pest-resistant plants.
- Higher food production to address global hunger and food security challenges.
- Customized nutrition crops, such as grains enriched with multiple vitamins and minerals.

3. Cancer Treatment Breakthroughs

- CRISPR-edited immune cells to precisely target and destroy cancer cells.
- Development of safer and more effective cancer therapies with minimal side effects.
- Possible prevention of cancer through early genetic corrections.

4. De-Extinction and Conservation

- Bringing back extinct species like woolly mammoths or passenger pigeons to restore ecosystems.
- Protecting endangered species by enhancing disease resistance and genetic diversity.
- Controlling invasive species and saving fragile ecosystems using gene drives.

5. Environmental Sustainability

- CRISPR-based microbes for cleaning up plastic waste, oil spills, and pollution.
- Sustainable biofuel production to reduce dependency on fossil fuels.
- Reducing the environmental impact of industries by engineering eco-friendly materials.

6. Human Space Exploration

- Genetic modifications for astronauts to resist space radiation and low-gravity effects.
- Crops genetically engineered to grow in Martian or space environments, supporting space missions.

9.CONCLUSION

CRISPR technology is one of the most significant breakthroughs in modern science, providing scientists with a powerful tool to edit genes with remarkable precision and efficiency. It has transformed fields such as medicine, agriculture, environmental conservation, and research, offering solutions to complex challenges like genetic disorders, crop improvement, and biodiversity protection. The technology's versatility allows it to be applied across a wide range of organisms, making it invaluable for both practical applications and scientific discovery. Despite its immense potential, CRISPR comes with important considerations, including ethical concerns, off-target effects, and regulatory challenges, which must be carefully addressed to ensure safe and responsible use. Overall, when applied thoughtfully and ethically, CRISPR holds the promise to revolutionize healthcare, enhance food security, support environmental conservation, and drive innovations in biotechnology. Its continued development and responsible application could fundamentally reshape the future of science, medicine, and sustainable development.

REFRENCES

[1] Barrangou R., et al., "CRISPR provides acquired resistance against viruses in prokaryotes," Science, 2007.

[2] Jinek M., Chylinski, K., Fonfara, I., Hauer, M., Doudna, J. A., and Charpentier, E. (2012). A programmable dual-RNA–guided DNA endonuclease in adaptive bacterial immunity. Science, 337(6096), 816–821.

[3] Wang, F., Wang, C., Liu, P., Lei, C., Hao, W., Gao, Y., & Zhao, K. (2016). Enhanced rice blast resistance by CRISPR/Cas9-targeted mutagenesis of the ERF transcription factor gene. Journal of Genetics and Genomics, 43(6), 345–348.

[4] Shi, J., Gao, H., Wang, H., Lafitte, H. R., Archibald, R. L., Yang, M., & Habben, J. E. (2017). ARGOS8 variants generated by CRISPR-Cas9 improve maize grain yield under field drought stress conditions. Plant Biotechnology Journal, 15(2), 207–216.

[5] Waltz, E. (2016). CRISPR-edited crops free to enter market, skip regulation. Nature Biotechnology, 34(6), 582.

[6] Frangoul, H., Altshuler, D., Cappellini, M. D., Chen, Y. S., Domm, J., Eustace, B. K., & Corbacioglu, S. (2021). CRISPR-Cas9 gene editing for sickle cell disease and β-thalassemia. New England Journal of Medicine, 384(3), 252–260.

[7] Stadtmauer, E. A., Fraietta, J. A., Davis, M. M., Cohen, A. D., Weber, K. L., Lancaster, E., & June, C. H. (2020). CRISPR-engineered T cells in patients with refractory cancer. Science, 367(6481), eaba7365.

[8] Shapiro, B. (2017). Pathways to de-extinction: How close can we get to resurrection of an extinct species? Functional Ecology, 31(5), 996–1002.